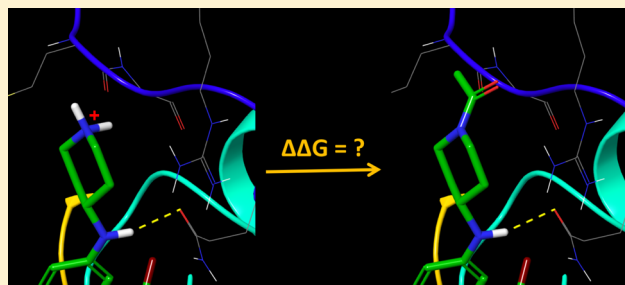


Accurate Calculation of Relative Binding Free Energies between Ligands with Different Net Charges

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ABSTRACT: In drug discovery programs, modifications that change the net charge of the ligands are often considered to improve the binding potency and solubility, or to address other ADME/Tox problems. Accurate calculation of the binding free-energy changes associated with charge-changing perturbations remains a great challenge of central importance in computational drug discovery. The finite size effects associated with periodic boundary condition and lattice summation employed in common molecular dynamics simulations introduce artifacts in the electrostatic potential energy calculations, which need to be carefully handled for accurate free-energy calculations between systems with different net charges. The salts in the buffer solution of experimental binding affinity assays also have a strong effect on the binding free energies between charged species, which further complicates the modeling of the charge-changing perturbations. Here, we extend our free-energy perturbation (FEP) algorithm, which has been extensively applied to many drug discovery programs for relative binding free-energy calculations between ligands with the same net charge (charge-conserving perturbation), to enable charge-changing perturbations. We have investigated three different approaches to correct the finite size effects and tested them on 10 protein targets and 31 charge-changing perturbations. We have found that all three methods are able to successfully eliminate the box-size dependence of calculated binding free energies associated with brute force FEP. Moreover, inclusion of salts matching the ionic strength of experimental buffer solution significantly improves the calculated binding free energies. For ligands with multiple possible protonation states, we applied the pK_a correction to account for the ionization equilibrium of the ligands and the results are significantly improved. Finally, the calculated binding free energies from these methods agree with each other, and also agree well with the experimental results. The root-mean-square error between the calculated binding free energies and experimental data is 1.1 kcal/mol, which is on par with the accuracy of charge-conserving perturbations. We anticipate that the outstanding accuracy demonstrated here across a broad range of target classes may have significant implications for drug discovery projects, where charge-changing modifications must be considered.



INTRODUCTION

The majority of small molecule drugs achieve their therapeutic effects through binding to a target protein, thereby either inhibiting or activating its biological functions. Therefore, accurate prediction of protein–ligand binding potency becomes a central task in computer-aided drug design.^{1–3} Alchemical free-energy perturbation (FEP) is one of the most rigorous and effective methods to calculate protein–ligand binding free energies.^{4–7} Previously, we have developed an alchemical free-energy calculation program called FEP+ for computing relative protein–ligand binding free energies.⁸ With many scientific and technological advances, including an improved molecular mechanics force field OPLS3,^{9,10} enhanced sampling by replica exchange with solute tempering,^{11,12} a highly efficient graphics processing unit implementation of molecular dynamics (MD) within the Desmond MD program,¹³ and advanced simulation setup, a high level of accuracy in the FEP+ calculated binding free energies (with a root-mean-square error (RMSE) of 1.1 kcal/mol) was achieved across a large number of protein targets and ligand series, in both retrospective validations and prospective applications.^{8,6} Recently, the FEP+ program was further extended to compute

relative binding free energies of scaffold hopping modifications (Core Hopping FEP), including ring opening/closing, ring extension, and ring size change perturbations.¹⁴ The Core Hopping FEP algorithm has also been further modified to calculate the relative binding free energies for macrocycles, which is a drug class containing large rings with a unique ability to balance various drug properties including solubility, permeability, and bioavailability.¹⁵

The FEP+ method mentioned above was limited to relative binding free-energy calculations between ligands with the same net charge (charge-conserving perturbation). However, in real drug discovery projects, modifications altering the net charges of compounds (charge-changing perturbation) are often employed to improve binding potency, ligand solubility, or other ADME/Tox properties. In particular, the introduction or removal of charged functional groups, such as ammonium or carboxylate moieties, are anticipated to greatly change the lipophilicity and lipophilic efficiency of molecules, and therefore are expected to have a significant impact on a wide

Received: August 7, 2018

Published: October 30, 2018

variety of ADME/Tox properties, perhaps most notably bioavailability and rate of clearance.¹⁶ Therefore, methods capable of predicting with high reliability the change in binding potency due to the introduction or removal of charged functional groups would be of great help to researchers attempting to maintain on-target activity while further optimizing the ADME/Tox profile of the project matter, which often is the principal challenge in small-molecule drug discovery.^{17,18}

Accurate prediction of the relative binding free energy between ligands with different net charges is difficult for several reasons. First, the periodic boundary condition (PBC) in commonly used MD simulations to reduce the interfacial effects of finite sized simulation box introduces artifacts in the electrostatic potential energy calculations. For a simulated system with a nonzero charge, the artificial interaction energy between the charged solutes in the central simulation box and their periodic images is non-negligible.^{19,20} The periodic images of charged solutes also perturb solvent structures in the central simulation box and cause artificial undersolvation.^{19,20} Furthermore, the commonly used lattice-summation methods such as particle-mesh Ewald (PME)²¹ to calculate electrostatic force in most MD programs, implicitly add a background plasma to neutralize the charged system, and artificially shifts the average of electrostatic potential in the simulation box to zero. This treatment in the lattice summation leads to inconsistency in the zero electrostatic potential for different systems, depending on the solute, the solvent, and their relative volumes in the simulation box, which results in systematic overestimation or underestimation of electrostatic potential energy for charged solutes.²² Second, the salts in the buffer solution used in experimental binding assays have a nontrivial effect on the binding affinity between charged ligands and proteins,²³ and these effects must be explicitly modeled for charge-changing perturbations to be accurately described. Third, ligands with different net charges usually have very different interactions with the protein receptor and the surrounding water molecules; thus, it may be very difficult to converge the free-energy calculations for such perturbations.

Previous efforts to model charged systems in free-energy calculations have mainly focused on eliminating the box size dependence in the electrostatic potential energy calculations. In 2013, Rocklin et al. introduced a method to perform post-simulation electrostatic potential energy correction to eliminate the box-size dependence of the charging free energy for charged systems.²⁰ In particular, the solvation free energy difference of the solute under nonperiodic boundary condition (NPBC), compared to that under PBC used in simulations, is suggested as a correction term added to the total potential energy. This correction term can be obtained approximately by numerically solving the Poisson–Boltzmann (PB) equation, or by analytical formulas.²⁰ To account for the difference between explicit solvent models used in simulations and implicit solvent models used in PB for solvation free-energy calculation, an additional correction term called “discrete solvent effects” is also introduced.²⁰ Lin et al. also proposed a similar method to correct the artifacts imposed by periodicity for charged systems.²² Both methods have the common correction for the solvation free energy difference between NPBC and PBC, with the major difference being how the discrete solvent effect is treated. Lin et al.’s method requires the calculation of the average of the electrostatic potential in the solvent phase and a correction term due to the shifting of the zero electrostatic

potential is suggested,²² where, in Rocklin’s method, this term is replaced by an analytical term derived from orientational disorder limit (ODL) of solvent molecules.²⁰ In parallel to these post-simulation correction methods, others have also considered introducing alchemical ions to neutralize the systems for charge-changing perturbations. In the work of Dixit and Chipot,²⁴ an sodium counterion was simultaneously mutated to a neutral particle as a charged ligand was annihilated for calculating the binding free energy of the charged ligand. Similarly, in constant pH MD simulations, the Shen group coupled neutralization of co-ions with protonation/deprotonation of titratable groups,²⁵ and later they replaced alchemical co-ions with alchemical hydroxide/hydroxonium.²⁶ Chen and Roux also utilized the conversion between an ion and a water molecule to maintain neutrality of a system.²⁷ The theory behind this type of methods is that, the dominant artifact in electrostatic potential energy calculation under PBC is proportional to the net charge of the solute, and the artifact is mostly eliminated when the system is neutral.

In this work, we investigated three approaches to calculate the relative binding free energies between ligands with different net charges. In the first method, we followed the work of Rocklin et al.²⁰ to perform post-simulation charge correction including correction terms for periodicity-induced net-charge interactions and undersolvation, residual integrated potential (RIP) effects, and discrete solvent effects; in the second method, similar to the work of Lin et al.,²² we added the correction for solvent phase electrostatic potential shifting during a MD simulation and the corrections for other periodicity-induced effects including net-charge interactions, undersolvation, and RIP effects at the end of the simulation; in the third method, we introduced co-alchemical ions similar to what has been used in constant pH MD simulations,^{25–27} including three variants: converting an ion to a dummy particle, a neutral particle, or a water molecule. In addition to the treatment of artifacts in electrostatic potential energy calculations, Na⁺ and Cl[−] ions matching the ionic strength of the buffer solution in experimental binding affinity assays were also modeled explicitly to investigate their effect on the calculated binding free energies. We tested these methods on a large set of protein targets and ligands with different net charges. The calculated binding free energies from the three approaches are within the statistical errors of each other, and also agree well with experimental values.

METHODS

Methods To Address Electrostatic Potential Energy for Charge-Changing Perturbations. For a perturbation between two ligands with different net charges (charge-changing perturbation), finite size effects due to PBC and lattice summation must be carefully considered.²⁰ To eliminate finite size effects in relative binding free-energy calculations for charge-changing perturbations, we have implemented the following three approaches, which we referred as charge FEP methods.

Post-simulation Charge Correction. With PBC and lattice summation employed in MD simulations, the charging free energy of a solute has errors due to the finite-sized simulation box. Following the work of Rocklin et al.,²⁰ the errors can be decomposed into the following four terms: periodicity-induced net-charge interactions for a point charge, periodicity-induced undersolvation for a point charge, the difference between a point charge and the real solute with partial charges distributed

on many atoms (also called the residual integrated potential, RIP, term), and the discrete solvent effects.²⁰ To correct the finite-size errors, we added three correction terms to the final free-energy difference between the two ligands subsequent to an uncorrected FEP simulation for a charge-changing perturbation $L_1 \rightarrow L_2$ (L_1 and L_2 are two ligands with different net charges):

$$\Delta G_{L_1 \rightarrow L_2}^{\text{corrected}} = \Delta G_{L_1 \rightarrow L_2}^{\text{original}} + \Delta G_{\text{NET-USV}} + \Delta G_{\text{RIP}} + \Delta G_{\text{DSC}} \quad (1)$$

The first correction term accounts for periodicity-induced net-charge interactions and undersolvation:²⁰

$$\Delta G_{\text{NET-USV}} = -\frac{\xi_{\text{LS}}}{8\pi\epsilon_0\epsilon_s} [Q_{S_2}^2 - Q_{S_1}^2] \frac{1}{L} \quad (2)$$

where Q_{S_1} and Q_{S_2} are the net charges of the simulated systems for ligands 1 and 2, respectively. These include charges from all particles of the simulated systems (ligands, protein, counterions, and salts, if included). L is the average edge length of the cubic simulation box used in the simulation, ξ_{LS} the cubic lattice-sum integration constant ($\xi_{\text{LS}} \approx -2.837297$), ϵ_0 the permittivity of a vacuum, and ϵ_s the solvent dielectric constant.

The second correction term (the RIP term) accounts for the difference between the solute with charge distributed on many atoms and the point charge assumed in the first correction term:²⁰

$$\Delta G_{\text{RIP}} = [(I_p + I_{L_2})Q_{S_2} - (I_p + I_{L_1})Q_{S_1}] \frac{1}{L^3} \quad (3)$$

where I_p is the RIP for protein, and I_{L_1} and I_{L_2} are the RIPs for ligands L_1 and L_2 , respectively. The RIP is calculated as the difference between the spatial integral of the electrostatic potential for a real solute in solvent and the integral of the electrostatic potential for a point charge of the same magnitude in solvent under NPBC by solving the Poisson–Boltzmann equation with APBS.²⁸ Note that RIP is calculated for solute atoms only and ions are not considered in the RIP calculations, because they are constantly moving around in the simulations in reference to the solute. In principle, the RIP term should be calculated for every frame in the trajectory and averaged in the final correction. In practice, since the RIP term is very small in magnitude and the variation of the term between different frames is also very small, we only used the last frame of a simulation for the RIP calculation.

The last correction term (the discrete solvent effects) accounts for the difference between explicit water molecules used in simulations and the implicit solvent models considered in previous two correction terms. Following the work of Rocklin et al.,²⁰ under ODL of solvent molecules, the discrete solvent effect has the following analytical form:

$$\Delta G_{\text{DSC}} = -\frac{\gamma_s N_s}{6\epsilon_0 L^3} [Q_{S_2} - Q_{S_1}] \quad (4)$$

where N_s is the number of solvent molecules in the simulation system and γ_s is the quadrupole-moment trace of a solvent molecule, relative to its single van der Waals interaction site. For a solvent model (e.g., SPC²⁹) with multiple interaction sites of charge q_i and distance r_i from the van der Waals interaction site, the quadrupole-moment trace has the following simple form:

$$\gamma_s = \sum_i q_i r_i^2 \quad (5)$$

These three correction terms are calculated for perturbations in the protein complex and in the solvent, respectively, and the difference of the corrected free energies in complex and solvent is the final estimated relative binding free energy between the two ligands.

Solvent Potential Correction. Lattice-summation methods artificially enforce the average of the electrostatic potential of a simulation box to be zero,²² and the zero electrostatic potential is thus defined differently for different systems, depending on the solute, the solvent, and their relative volumes in the simulation box. In the charge correction method mentioned above, the error in the electrostatic potential energy due to the shift of zero electrostatic potential is calculated analytically under ODL of the solvent (the discrete solvent effect term). As suggested by Lin et al.,²² we can also explicitly calculate the average of the electrostatic potential of the solvent region, $\langle\phi\rangle_s$, and the error in the electrostatic potential energy for a system with a net charge of q can be simply corrected by the following formula:

$$U_{\text{corr}} = -\langle\phi\rangle_s q \quad (6)$$

We calculated the average of the electrostatic potential of all grid points in the solvent region in a MD simulation, and have added the correction energy directly to the total potential during the simulation. The other two correction terms for the periodicity induced net-charge interactions and undersolvation, and RIP effects (eqs 2 and 3) are also added after the simulation similarly as is performed in the post-simulation charge correction method.

Co-alchemical Ion. The dominant errors arising from PBC and lattice summation are proportional to the net charge of the simulated system.²⁰ If the net charge of the system is zero for both physical end points, the finite size effects are negligible. Following charge-leveling techniques used in constant pH MD methods,^{25–27} we have introduced co-alchemical ion(s) (either sodium or chloride) into simulations of charge-changing perturbations to achieve a charge-conserving simulation protocol. In particular, for a perturbation between a neutral ligand and a charged ligand, a special ion denoted as a “co-alchemical ion” is computed as the ligand is mutated, keeping the net charge of the simulated system to be zero for both end points. Three different variations of the co-alchemical ion approach were used. In the first variant, termed the “dummy co-alchemical ion approach”, a co-alchemical ion is mutated into a dummy particle (a point mass without charge and Lennard-Jones interactions). During the transformation from the ion to the dummy particle, the charge is first reduced to zero, then the Lennard-Jones interactions are turned off. Also, the soft-core scheme is used for the Lennard-Jones interactions to avoid singularity.³⁰ In the second variant termed the neutral co-alchemical ion approach, a co-alchemical ion is converted to a neutral particle with the same Lennard-Jones interactions as the ion but zero charge. In the third variant termed the co-alchemical water approach, an alchemical ion is perturbed to a water molecule: in the water state, the O and H atoms have the same partial charges and Lennard-Jones parameters as a regular water; in the ion state, the charge and Lennard-Jones parameters of the O atom are changed to those of the ion while the two H atoms become dummy particles without

charges and Lennard-Jones interactions. In all of the variations, the co-alchemical ion is free to move in the simulation box.

pK_a Correction. For a ligand with multiple non-negligible protonation states present in solvent and/or in protein complexes, we applied the pK_a correction method (details have been presented in a companion paper)³¹ to account for all possible protonation states. In summary, for a pair of ligands that both have two protonation states, we first calculated the relative binding free energies between the deprotonated forms of the two ligands, and also the relative binding free energy between the deprotonated and the protonated forms of each ligand. The final relative binding free energy between the two ligands is then calculated as

$$\Delta\Delta G_b(\text{pH}) = \Delta\Delta G_b^0 - RT \ln \left(\frac{1 + 10^{\text{pK}_{a,2}^{\text{C}} - \text{pH}}}{1 + 10^{\text{pK}_{a,2}^{\text{S}} - \text{pH}}} \right) + RT \ln \left(\frac{1 + 10^{\text{pK}_{a,1}^{\text{C}} - \text{pH}}}{1 + 10^{\text{pK}_{a,1}^{\text{S}} - \text{pH}}} \right) \quad (7)$$

$\Delta\Delta G_b^0$ is the relative binding free energy between the two ligands when the deprotonated forms of the ligands are used. $\text{pK}_{a,i}^{\text{S}}$ and $\text{pK}_{a,i}^{\text{C}}$ are the pK_a values of ligand *i* in the solvent and complex, respectively. $\text{pK}_{a,i}^{\text{S}}$ is obtained from experimental measurements or from quantum mechanics calculation, by using the Jaguar program,³² while $\text{pK}_{a,i}^{\text{C}}$ is calculated according to

$$\text{pK}_{a,i}^{\text{C}} = \text{pK}_{a,i}^{\text{S}} - \frac{\Delta\Delta G_{b,i}(0 \rightarrow +)}{RT \ln 10} \quad (8)$$

where $\Delta\Delta G_{b,i}(0 \rightarrow +)$ is the relative binding free energy between the deprotonated and protonated forms of the ligand. In cases where a single protonation state is dominant for one of the ligands, its corresponding correction term in eq 7 is removed.

Simulation Protocol. We have implemented the above charge FEP methods in the FEP+ program⁸ of Schrödinger Suite 2018,³³ which was used for the relative binding free-energy calculations reported here. We tested these methods on 10 protein targets, including CDK2,³⁴ DLK,³⁵ EGFR,³⁶ EPHX2,³⁷ IRAK4,³⁸ ITK,³⁹ JAK1,^{40,41} JNK1,⁴² PTP1B,⁴³ and TYK2^{44,45} (see Table 1). The initial structures of the targets were taken from the Protein Databank (PDB) with PDB IDs 1OIY, 5CEQ, 3LZB, 3KOO, 4ZTM, 4ZTN, 4PQN, 4IVD, 2GMX, 2QBP, and 4GIH. The proteins were prepared with Protein Preparation Wizard⁴⁶ and ligands without crystal structures were built based on binding mode of the most similar crystal ligand.⁴⁷ A total of 47 ligands were included, 21 of which have a perturbed charged moiety. For these 21 ligands, both protonated and deprotonated forms were included to account for possible change of protonation states upon binding. For each target and its associated ligands, we used the FEP+ GUI^{8,33} to build a perturbation map, which includes multiple perturbations in closed cycles. In total, 111 perturbations were calculated, including three types of perturbations: 52 charge-changing perturbations, 8 charge-hopping perturbations between ligands with the same net charges but with a charged group perturbed to a different charged group or to a different position in the ligand, and 51 neutral perturbations that have neither net charge change nor charged group perturbed. Excluding charge-changing perturbations between protonated and deprotonated forms of the same

Table 1. Simulation Systems^a

protein ^b	PDB	exp. pH	ionic strength (mM)	No. of ligands	No. of charge pert.
CDK2	1OIY	7.5	56	3	2
DLK	5CEQ	7.5	54	5	2
EGFR	3LZB	7.5	60	5	4
EPHX2	3KOO	7.0	206	4	2
IRAK4	4ZTM/ 4ZTN	7.2	21	9	8
ITK	4PQN	7.2	61	4	2
JAK1	4IVD	7.2	61	6	2
JNK1	2GMX	7.2	40	3	3
PTP1B	2QBP	7.0	150	3	2
TYK2	4GIH	7.0	150	5	4
total:				47	31

^aIonic strength is estimated according to experimental buffer condition (no information for TYK2 was found, so pH 7 and an ionic strength of 150 mM were used). ^bLegend: CDK2, cyclin-dependent kinases 2;³⁴ DLK, dual leucine zipper kinase;³⁵ EGFR, epidermal growth factor receptor;³⁶ EPHX2, soluble epoxide hydrolase;³⁷ IRAK4, interleukin-1 receptor associated kinase-4;³⁸ ITK, interleukin-2 inducible T-cell kinase;³⁹ JAK1, Janus kinase 1;^{40,41} JNK1, c-Jun N-terminal kinase 1;⁴² PTP1B, protein tyrosine phosphatase 1B;⁴³ and TYK2, tyrosine kinase 2.^{44,45}

ligand, there are 31 charge-changing perturbations between different ligands.

For each perturbation, the solute was soaked in a cubic SPC²⁹ water box with an 8 Å buffer width for the complex perturbation leg and a 10 Å buffer width for the solvent perturbation leg. OPLS3 force field was used for proteins and ligands.¹⁰ For charge-changing and charge-hopping perturbations, the system was neutralized with Na⁺ or Cl[−] ions and additional Na⁺/Cl[−] salts matching the ionic strength of the experimental assay condition were also included. Since ligands with different net charges or different charged groups usually interact differently with the protein receptors and surrounding water molecules, the number of λ windows for charge-changing and charge-hopping perturbations were increased to 24 from 12 for neutral perturbations. Each λ window was equilibrated independently with the default protocol in FEP+.⁸ Production simulations were performed with the solute tempering replica exchange method.^{11,12} The atoms in the REST region were selected through the automated REST region assignment algorithm implemented in FEP+.⁸ The temperature ladder for REST was generated based on the number of atoms in the REST region, the number of replicas, and the target exchange probability of 0.3, following the algorithm of Patriksson and van der Spoel.⁴⁸ Replica exchanges between neighboring λ windows were attempted every 1.2 ps. The production runs lasted 30 ns per λ window (50 ns for EPHX2, because of slow convergence). For each of the three charge FEP approaches mentioned above, the simulations were repeated three times with different random seeds, to assess the convergence errors. Free energies between neighboring λ windows were calculated using the Bennett acceptance ratio method.⁴⁹ Cycle closure correction^{50,51} and pK_a correction (see above) were applied to relative binding free energies when appropriate. The data for uncorrected FEP calculations were simply obtained from FEP simulations prior to the post-simulation charge correction.

To evaluate whether the three approaches effectively eliminated the effects of the finite size on the calculated

binding free energies, we investigated the box size dependence of the calculated binding free energy for three charge-changing perturbations (one for each of DLK, PTP1B, and CDK2). Four different-sized simulation boxes were tested with buffer widths of 8, 10, 15, and 20 Å, respectively. To investigate the effects of the ionic strength of the buffer solution on the calculated binding free energies, we performed two sets of simulations: one with only counterions to neutralize the system and the other with additional salts to mimic the buffer solution used in the experimental assay (150 mM for DLK and PTP1B and 56 mM for CDK2). Each simulation was run for 30 ns per λ window and was repeated three times with different random seeds to assess the statistical uncertainty. The accumulated simulation time of this work is ~ 1.1 ms.

RESULTS

Eliminating Box Size Dependence of Relative Binding Free Energy. Because of the finite-size effects associated with PBC and lattice summation, the calculated charging free energy and the resulting binding free energy of a charged ligand from brute force FEP simulations are dependent on the size of the simulation box.²⁰ To test whether the three approaches introduced above can effectively remove the box size dependence, we performed FEP simulations for charge-changing perturbations with four different box sizes: 8, 10, 15, and 20 Å water buffer. Three charge-changing perturbations were tested, each with a different protein target. The first one is a perturbation between a positively charged piperidine group (ligand 26) and a neutral 1-acetylpiperidine group (ligand 27) binding to protein tyrosine phosphatase 1B (PTP1B)⁴³ (see Figure 1A). The experimental data indicates that ligand 27 is 1.8 kcal/mol more potent than ligand 26. In the complex structure, the perturbed group is in close proximity to two arginine residues, Arg24/254 (Figure 1B). The repulsive interactions between Arg24/254 and the charged piperidine group of ligand 26 might be responsible for the lower potency of ligand 26, compared to ligand 27. We first performed FEP simulations with counterions only. For the uncorrected FEP neglecting special treatment of electrostatic potential energy, the relative binding free energy gradually become more negative when the simulation box size is increased (black lines in Figure 1C): it is reduced from -1.7 kcal/mol to -3.3 kcal/mol with the increase of the buffer size from 8 Å to 20 Å. In contrast, when any of the three charge FEP methods is applied, the calculated relative binding free energies are essentially independent of the box size with a value of approximately -3.5 kcal/mol (Figure 1C). The results from the three methods agree well with each other, but differ significantly from the results of the uncorrected FEP. This indicates that the artifacts arising from PBC and lattice summation in the uncorrected FEP are non-negligible, can cause strong box size dependence in the calculated binding free energy, and were effectively eliminated by the three charge FEP implementations reported here.

Although the results from the counterions-only simulations of the three charge FEP methods have negligible dependence on the size of the simulation box, the calculated relative binding free energy is ~ 2 kcal/mol lower than the experimental value. We then performed another set of FEP simulations with 150 mM salts in addition to the counterions to investigate the effect of the ionic strength of the buffer solution on the calculated binding free energy. Interestingly, the salts have a very strong effect on the calculated binding free

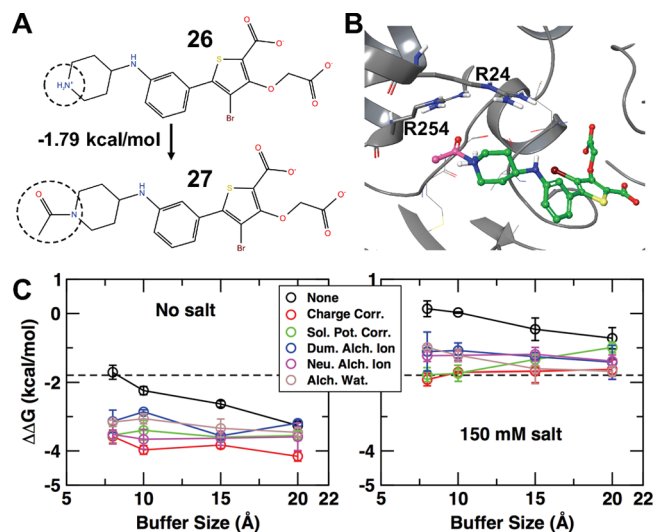


Figure 1. Charge FEP methods eliminate box size dependence of relative binding free energy for charge-changing perturbation of PTP1B ligands.⁴³ (A) Perturbation between ligand 26 (with -1 charge) and ligand 27 (with -2 charge) binding to PTP1B. The perturbed groups are highlighted by dashed circles. (B) Structures of protein–ligand binding complexes. Ligands 26 and 27 are shown in green and pink, respectively. Note that the common parts of the ligands overlap each other, so only one color can be seen. Arg24 and Arg254 in the PTP1B binding pocket are also shown. (C) FEP results without salts (left) or with 150 mM salt (right) under four different simulation box sizes (8, 10, 15, and 20 Å water buffer). Solid lines are results from different charge FEP protocols: black, red, green, blue, magenta, and brown lines represent uncorrected FEP, FEP with post-simulation charge correction, FEP with solvent potential correction, FEP with dummy co-alchemical ion, FEP with neutral co-alchemical ion, and FEP with co-alchemical water, respectively. Error bars indicate standard deviations of three simulations with different random seeds. The horizontal dashed line represents the experimental value.

energy. Compared to the results with counterions only, the results with 150 mM salt are all shifted up (i.e., more positive) by ~ 2 kcal/mol (Figure 1C). This is probably due to the ion-mediated interaction between the protein and charged ligand. In the simulations with addition of salts, a Cl^- ion was observed to penetrate into the region between the piperidine group of ligand 26 and two arginines in the binding pocket, reducing the repulsive interactions between ligand 26 and Arg24/254 (Figure 1B). This does not occur in the counterions-only simulations, because the complex has a negative net charge and only Na^+ ions are added to neutralize the system. The calculated relative binding free energy from the uncorrected FEP including the added salts is still box-size dependent, decreasing from 0.1 kcal/mol to -0.7 kcal/mol with the increase of the buffer size from 8 Å to 20 Å. In comparison, the results from the three charge FEP methods are essentially independent of the size of the simulation box, agree well with the experimental value (~ 1.4 in calculation versus 1.8 in experiment), and further agree with each other.

The second charge-changing perturbation involves a neutral 1-acetylpiperidine group (ligand 9) that is transformed to a positively charged piperidine group (ligand 4) for a ligand binding to dual leucine zipper kinase (DLK)³⁵ (Figure 2A). The perturbation region is mostly solvent exposed in the protein complex (Figure 2B). According to the experimental data, ligand 4 is 0.6 kcal/mol more potent than ligand 9. The

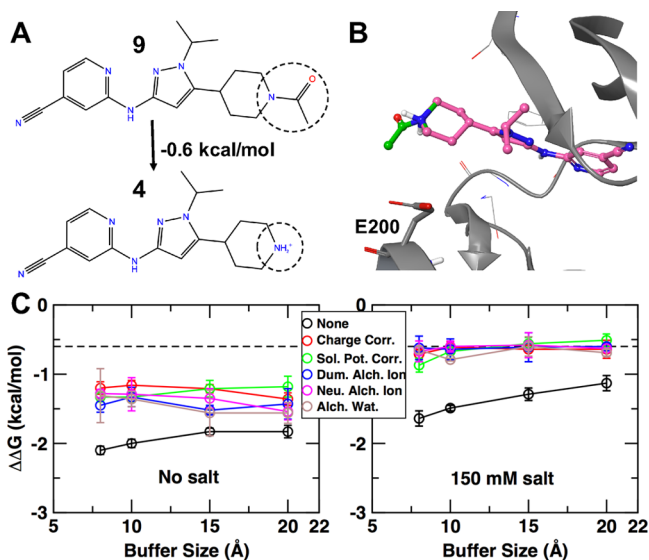


Figure 2. Charge FEP methods eliminate box size dependence of relative binding free energy for charge-changing perturbation of DLK ligands.³⁵ (A) Perturbation between neutral ligand 9 and positively charged ligand 4 binding to DLK. The perturbed groups are highlighted by dashed circles. (B) Structures of protein–ligand complexes. Ligands 9 and 4 are shown in green and pink, respectively. Note that the common parts of the ligands overlap each other, so only one color can be seen. Glu200 in the DLK binding pocket is also shown. (C) FEP results without salts (left) or with 150 mM salt (right) under four different simulation box sizes (8, 10, 15, and 20 Å water buffer). Solid lines are results from different charge FEP protocols: black, red, green, blue, magenta, and brown lines represent uncorrected FEP, FEP with post-simulation charge correction, FEP with solvent potential correction, FEP with dummy co-alchemical ion, FEP with neutral co-alchemical ion, and FEP with co-alchemical water, respectively. Error bars indicate standard deviations of three simulations with different random seeds. The horizontal dashed line represents the experimental value.

charged amine of ligand 4 was observed to interact with the negatively charged Glu200 in some frames of the simulation (Figure 2B), which might be the reason why ligand 4 has a higher potency. We first performed a set of FEP simulations with counterions only. For the uncorrected FEP, the calculated relative binding free energy between the two ligands is approximately -2 kcal/mol for a simulation box of 8 Å buffer width, and the calculated free energy increases from -2.1 kcal/mol to -1.8 kcal/mol with the increase of the simulation box size from 8 Å to 20 Å (Figure 2C). By comparison, the calculated relative binding energies from any of the three charge FEP approaches are approximately the same for simulations of different box sizes with a value of approximately -1.4 kcal/mol (Figure 2C). While the results from the three methods are similar to each other, they are significantly different from the results of the uncorrected FEP. This again indicates the artifacts associated with PBC and lattice summation in the uncorrected FEP are non-negligible and the three charge FEP approaches have eliminated these artifacts.

While the results from the three charge FEP approaches from the counterions-only simulations are consistent with each other and are significantly improved, compared with uncorrected FEP, there is still some difference between the calculated binding free energy and the experimental result. As above, we performed another set of FEP simulations with 150

mM salts in addition to the counterions to investigate the effect of ionic strength. Similarly, while the calculated relative binding free energy from the uncorrected FEP still has a dependence on the simulation box size (increased from -1.6 kcal/mol to -1.1 kcal/mol when the buffer size is increased from 8 Å to 20 Å (Figure 2C)), the three charge FEP approaches effectively removed the box size dependence and results from three methods agree with each other. Furthermore, the salts from the buffer solution have a strong effect on the calculated binding free energy. Compared to the simulations with counterions only, the results from simulations with additional salts were shifted upward (i.e., more positive), by ~ 0.5 kcal/mol, to a value of -0.7 kcal/mol, in good agreement with the experimental value. This might be due to the salt screening effect, which weakens the interaction between ligand 4 and Glu200.

The third charge-changing perturbation involved the transformation of a negatively charged carboxylate group (ligand 39) to a neutral hydroxyl group (ligand 25) for a ligand binding to cyclin-dependent kinase 2 (CDK2)³⁴ (Figure 3A). The experimental data indicated ligand 25 is 1.5 kcal/mol more potent than ligand 39. In the complex structure, the perturbed group is close to two charged residues, Asp86 and Lys89 (Figure 3B). In the simulations, we observed that the hydroxyl group of ligand 25 sometimes formed a hydrogen bond with Asp86, while the carboxylate group of ligand 39

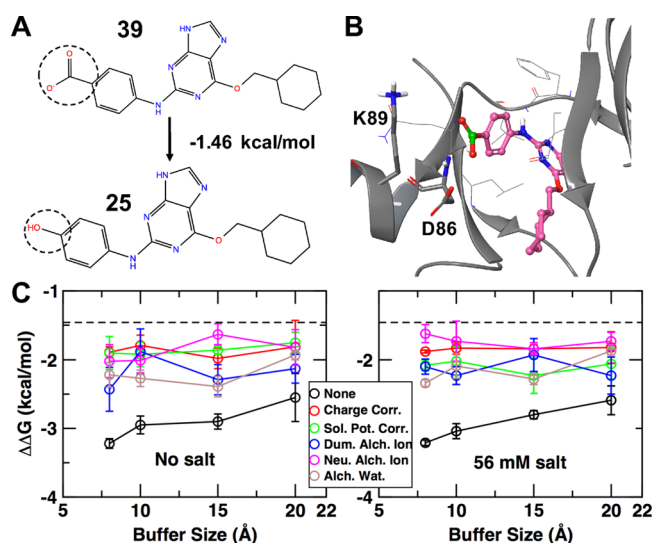


Figure 3. Charge FEP methods eliminate box size dependence of relative binding free energy for charge-changing perturbation of CDK2 ligands.³⁴ (A) Perturbation between negatively charged ligand 39 and neutral ligand 25 for CDK2. The perturbed groups are highlighted by the dashed circles. (B) Structures of protein–ligand complexes. Ligands 39 and 25 are shown in green and pink, respectively. Note that the common parts of the ligands overlap each other, so only one color can be seen. Asp86 and Lys89 in the CDK2 binding pocket are also shown. (C) FEP results without salts (left) or with 56 mM salt (right) under four different simulation box sizes (8, 10, 15, and 20 Å water buffer). Solid lines are results for different charge FEP protocols: black, red, green, blue, magenta, and brown lines represent uncorrected FEP, FEP with postsimulation charge correction, FEP with solvent potential correction, FEP with dummy co-alchemical ion, FEP with neutral co-alchemical ion, and FEP with co-alchemical water, respectively. Error bars indicate standard deviations of three simulations with different random seeds. The horizontal dashed line represents the experimental value.

sometimes formed a salt bridge with Lys89 but unfavorably interacted with Asp86. This might be the reason that ligand 25 is more potent than ligand 39. As previously noted, we first performed a set of FEP simulations with counterions only. For the uncorrected FEP, the calculated relative binding free energy is clearly box-size dependent. It is increased from -3.2 kcal/mol to -2.6 kcal/mol when the buffer size is increased from $8 \text{ \AA}$ to $20 \text{ \AA}$ (Figure 3C). However, with the charge FEP methods, the calculated relative binding free energy is essentially independent of the size of the simulation box and is also shifted upward to approximately -2 kcal/mol, closer to the experimental value (Figure 3C). The results from different charge FEP methods are consistent with each other.

We then performed another set of FEP simulations with 56 mM salts in addition to counterions, with same ionic strength as the experimental buffer solution, to investigate the salt effect. In contrast to the cases of DLK and PTP1B, the calculated relative binding free energy for this pair of CDK2 ligands is not significantly affected by the addition of salts (Figure 3C). The results are similar between the simulations with counterions only and the simulations with 56 mM salt. To check whether the insensitivity of the free-energy results, with respect to the addition of salts is due to the low salt concentration used, we also tried to increase the salts concentration to 150 mM and similar results were obtained (data not shown). In the complex, the two residues near the perturbed group of the ligands, Asp86 and Lys89 (Figure 3B), have opposite charges. The salt screening effects on the interactions of these two residues with the ligands might compensate each other, which might be the reason why the binding of the ligands are insensitive to the ionic strength.

Accurate Calculation of Relative Binding Free Energy for CDK2 Ligands. After confirming that the three charge FEP methods effectively eliminate the artifacts from PBC and lattice summation, and give correct relative binding free energies of charge-changing perturbations, we systematically tested the charge FEP methods on multiple ligands and multiple protein targets (recall Table 1). Since the relative binding free energy has negligible dependence on the simulation box size with the charge FEP methods (Figures 1–3), we used a simulation box with $8 \text{ \AA}$ water buffer for these simulations. Counterions and additional salts with concentrations matching the appropriate experimental buffer solutions were added in the following simulations.

As an example, we tested the charge FEP for three ligands binding to CDK2³⁴ (Figure 4). Ligand 39 is negatively charged due to a carboxylate group, while ligands 25 and 34 are neutral with hydroxyl and amide groups, respectively. Three perturbations were run forming a closed cycle. Two of the perturbations are charge-changing perturbations ($39 \rightarrow 25$ and $39 \rightarrow 34$), while the third one is a neutral perturbation ($34 \rightarrow 25$). The results from the three charge FEP methods agree well with each other with a maximum difference of <0.7 kcal/mol, and are also in agreement with experimental results (see Table 2). Utilizing the co-alchemical water approach, the errors of calculated relative binding free energies, compared with experimental values for the two charge-changing perturbations are ~ 0.6 kcal/mol (Table 2 and Figure 4). For the uncorrected FEP, the errors of calculation are large— 1.8 kcal/mol error for perturbation from ligand 39 to ligand 25 and 0.9 kcal/mol error from ligand 39 to ligand 34—resulting in a RMSE of ~ 1.5 kcal/mol (see Table 2). In contrast, when any charge FEP method is applied, calculation errors are significantly

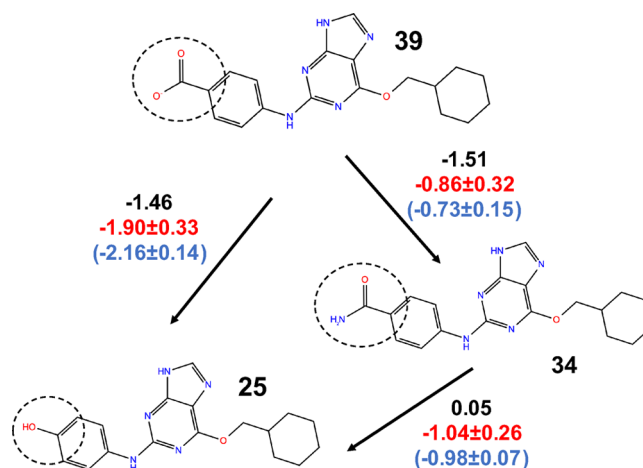


Figure 4. Charge-changing perturbations among CDK2 ligands.³⁴ Ligand 39 is negatively charged, while ligands 25 and 34 are neutral. The perturbed groups are highlighted by the dashed circles. Black numbers are experimental relative binding free energies. Calculated relative binding free energies are from the first trial of three independent FEP simulations with co-alchemical water. Blue and red numbers represent calculated relative binding free energies before and after cycle closure correction, respectively. The binding free energy is given in units of kcal/mol.

Table 2. Calculated Binding Free Energies for Charge-Changing Perturbations of CDK2 Ligands^a

scheme ^b	Binding Free Energy (kcal/mol)		RMSE ^c
	perturbation 39 $\rightarrow$ 25	perturbation 39 $\rightarrow$ 34	
exp.	-1.46	-1.51	
none	-3.29	-2.43	1.45
charge corr.	-1.87	-1.00	0.46
sol. pot. corr.	-2.18	-1.55	0.51
dum. alch. ion	-2.13	-1.42	0.48
neu. alch. ion	-2.09	-1.18	0.50
alch. wat.	-1.90	-0.86	0.56

^aThe experimental values are taken from ref 34. The calculated values (after cycle closure correction) are from the first trial of three sets of FEP simulations. ^bLegend: none, uncorrected FEP; charge corr., FEP with post-simulation charge correction; sol. pot. corr., FEP with solvent potential correction; dum. alch. ion, FEP with dummy co-alchemical ion; neu. alch. ion, FEP with neutral co-alchemical ion; and alch. wat., FEP with co-alchemical water. ^cRMSE = root-mean-square error from experimental values.

reduced. The error for perturbation from ligand 39 to ligand 25 is decreased to 0.4 – 0.7 kcal/mol, while the error for perturbation from ligand 39 to ligand 34 is reduced to 0 – 0.7 kcal/mol. The overall RMSE is reduced to 0.5 kcal/mol.

Accurate Calculation of Relative Binding Free Energy for IRAK4 Ligands with pK_a Correction. In the above example of CDK2, ligand 39 is predominantly charged and does not change its protonation state upon binding to the protein. However, in many of the cases tested here, the pK_a value of a ligand is close to the experimental pH, thus multiple protonation states may contribute to the binding, or the protonation state of the ligand may change upon binding. In such cases, we must apply the pK_a correction method introduced above to account for the contributions of all accessible protonation states.

For example, we tested five ligands binding to interleukin-1 receptor associated kinase-4 (IRAK4)³⁸ (Figure 5). Ligand 5

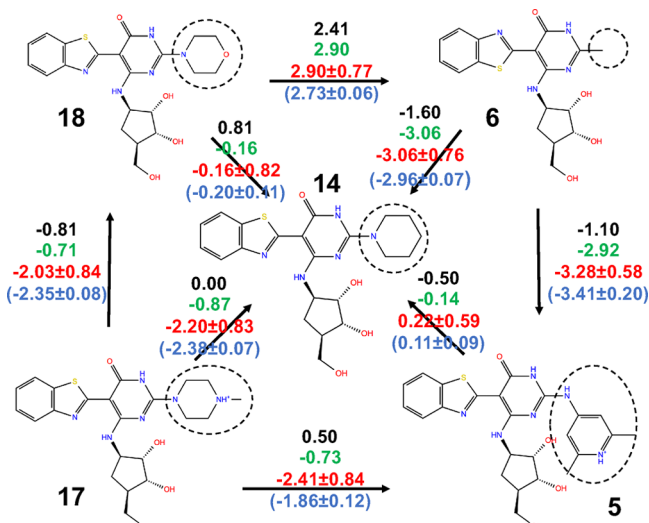


Figure 5. Charge-changing perturbations of IRAK4.³⁸ Ligands 5 and 17 are positively charged, while ligands 6, 14, and 18 are neutral. The perturbed groups are highlighted by the dashed circles. Black numbers are experimental relative binding free energies. Calculated relative binding free energies are from the first trial of three independent FEP simulations with co-alchemical water. Blue and red numbers represent calculated relative binding free energies before and after cycle closure correction, respectively, while green numbers are calculated binding free energies after applying pK_a correction. The unit of binding free energy is kcal/mol.

has a 2,6-dimethylpyridine group, while ligand 17 has a 1-methylpiperazine group. The pK_a values of ligands 5 and 17 in solvent are 7.2 and 7.7, respectively, according to quantum mechanics calculation with Jaguar;³² these values are close to the experimental pH of 7.2. Therefore, the ligands have mixed populations of charged and neutral forms in solvent. The other three ligands (ligands 6, 14, and 18) are neutral. Eight perturbations were run, forming closed thermodynamic cycles

in the perturbation map (see Figure 5). We first modeled the charged forms of ligands 5 and 17, with four charge-changing perturbations ($6 \rightarrow 5$, $5 \rightarrow 14$, $17 \rightarrow 14$, and $17 \rightarrow 18$), one charge-hopping perturbation ($17 \rightarrow 5$), and three neutral perturbations ($6 \rightarrow 14$, $18 \rightarrow 6$, and $18 \rightarrow 14$). The results of the calculations, along with the experimental values, are presented in Table 3. With uncorrected FEP, the calculation results have large errors compared with the experiment, and a RMSE of 2.4 kcal/mol. With the charge FEP methods, the calculation is improved with the RMSE reduced to 1.8–2.2 kcal/mol. The calculation errors are mainly from perturbations involving ligand 17. One possible reason is that the mixed protonation forms of ligand 17 was not considered, and the dominant protonation state might be different in complex versus in solvent. Therefore, we applied pK_a correction (see the Methods section for details) to account for both protonation states. From the FEP simulations between charged and neutral forms of ligands 5 and 17, the pK_a of ligand 5 is estimated to be shifted up to ~ 8 in complex while the pK_a of 17 is estimated to be reduced to ~ 6 . Therefore, in complex, the dominant form of ligand 5 is charged while the dominant form of ligand 17 is neutral. Based on the calculated pK_a values in complex and solvent, we added the pK_a correction term to the original binding free energies. The pK_a correction significantly improves the results on top of the charge FEP methods, with the RMSE further reduced to 0.9–1.2 kcal/mol (Table 3).

Large-Scale Testing of Charge FEP Methods. To systematically test the charge FEP methods, we have assembled a set of 10 protein targets (Table 1), each with 3–9 congeneric ligands with different net charges (47 ligands in total). Perturbations forming closed cycles as shown for CDK2 and IRAK4 were run for these ligands (Figures 4 and 5). These perturbations consist of charge-changing, charge-hopping, and neutral perturbations. The total number of charge-changing perturbations is 31. To account for possible protonation state change, we included both protonated and deprotonated forms

Table 3. Calculated Binding Free Energies for Charge-Changing Perturbations of IRAK4 Ligands^a

scheme ^b	Binding Free Energy (kcal/mol)					RMSE ^c
	perturbation 5 → 14	perturbation 6 → 5	perturbation 17 → 14	perturbation 17 → 18	perturbation 17 → 5	
Before pK _a Correction						
exp.	−0.50	−1.10	0.00	−0.81	0.50	
none	1.90	−4.67	−0.78	−0.75	−2.68	2.42
charge corr.	−0.05	−2.77	−2.76	−2.73	−2.71	2.22
sol. pot. corr.	0.94	−3.13	−1.73	−2.05	−2.67	2.04
dum. alch. ion	0.37	−3.20	−2.10	−1.88	−2.48	1.98
neu. alch. ion	−0.09	−2.81	−2.24	−2.00	−2.15	1.82
alch. wat.	0.22	−3.28	−2.20	−2.03	−2.41	2.00
After pK _a Correction						
exp.	−0.50	−1.10	0.00	−0.81	0.50	
none	1.52	−4.29	−0.58	−0.55	−2.10	2.07
charge corr.	−0.33	−2.49	−0.93	−0.91	−0.61	0.90
sol. pot. corr.	0.60	−2.79	−0.58	−0.90	−1.18	1.20
dum. alch. ion	0.03	−2.85	−0.91	−0.69	−0.93	1.12
neu. alch. ion	−0.41	−2.50	−0.91	−0.66	−0.50	0.87
alch. wat.	−0.14	−2.92	−0.87	−0.71	−0.73	1.07

^aThe experimental values are taken from ref 38. The calculated values (after cycle closure correction) are from the first trial of three sets of FEP simulations. ^bLegend: none, uncorrected FEP; charge corr., FEP with post-simulation charge correction; sol. pot. corr., FEP with solvent potential correction; dum. alch. ion, FEP with dummy co-alchemical ion; neu. alch. ion, FEP with neutral co-alchemical ion; alch. wat., co-alchemical water.

^cRMSE: root-mean-square error from experimental values.

for ligands that have a titratable site being perturbed. We then performed pK_a correction for these ligands same as the IRAK4 example shown above (Table 3). In the following discussion, we only present the results after pK_a correction.

The FEP simulations for the entire test set were repeated three times with different random seeds. The RMSEs of the calculated relative binding free energies of the three runs are summarized in Table 4. For the uncorrected FEP, the mean

Table 4. RMSE of Calculated Relative Binding Free Energies for the Entire Dataset of 31 Charge-Changing Perturbations^a

scheme ^b	Binding Free Energy (kcal/mol)			
	Run 1	Run 2	Run 3	mean $\pm$ st dev ^c
none	1.48	1.43	1.49	1.47 $\pm$ 0.03
charge corr.	0.94	1.02	1.08	1.01 $\pm$ 0.07
sol. pot. corr.	1.03	1.02	1.08	1.04 $\pm$ 0.03
dum. alch. ion	1.02	1.05	1.03	1.03 $\pm$ 0.02
neu. alch. ion	0.98	1.01	1.12	1.04 $\pm$ 0.07
alch. wat.	1.08	1.07	1.05	1.07 $\pm$ 0.02

^aRuns 1–3 are independent FEP simulations with different random seeds. ^bLegend: none, uncorrected FEP; charge corr., FEP with postsimulation charge correction; sol. pot. corr., FEP with solvent potential correction; dum. alch. ion, FEP with dummy co-alchemical ion; neu. alch. ion, FEP with neutral co-alchemical ion; alch. wat., FEP with co-alchemical water. Runs 1–3 are independent FEP simulations with different random seed. ^cThe last column shows the mean and standard deviation of the root-mean-square error (RMSE) from experimental values of three runs.

RMSE of three runs is 1.47 kcal/mol with a very small standard deviation of 0.03 kcal/mol. By comparison, for the charge FEP methods, the mean RMSE is significantly reduced, ranging between 1.01 and 1.07 for various methods, and the variations between different charge FEP methods are within statistical error, indicating the slightly different RMSEs among these methods are not statistically significant. However, we noticed that Bennett errors of the dummy co-alchemical ion method were larger than that of the other methods (mean value of 0.2 vs 0.1 kcal/mol for other methods) (see Table 5), indicating slower convergence of the dummy co-alchemical ion. The slow convergence of the dummy co-alchemical ion

Table 5. Bennett Errors of FEP Calculations for Charge-Changing Perturbations^a

scheme ^b	Bennett Error		
	Run 1	Run 2	Run 3
none	0.10 (0.31)	0.10 (0.24)	0.10 (0.26)
charge corr.	0.10 (0.31)	0.10 (0.24)	0.10 (0.26)
sol. pot. corr.	0.10 (0.25)	0.10 (0.26)	0.09 (0.27)
dum. alch. ion	0.24 (0.56)	0.22 (0.57)	0.23 (0.81)
neu. alch. ion	0.11 (0.25)	0.11 (0.26)	0.11 (0.26)
alch. wat.	0.16 (0.35)	0.16 (0.46)	0.16 (0.35)

^aMean (maximal) values of Bennett errors of all 52 charge-changing perturbations including perturbations between deprotonated and protonated forms of the same ligand. Runs 1–3 are independent simulations with different random seeds. ^bLegend: none, uncorrected FEP; charge corr., FEP with post-simulation charge correction; sol. pot. corr., FEP with solvent potential correction; dum. alch. ion, FEP with dummy co-alchemical ion; neu. alch. ion, FEP with neutral co-alchemical ion; and alch. wat., FEP with co-alchemical water.

approach might be due to the following reason. In the intermediate λ windows of some of the simulations with the dummy co-alchemical ion approach, when the charge of the co-alchemical ion is zero, it was observed to enter the hydrophobic regions of protein interior and remain there in the rest of the trajectories. When the charge of the co-alchemical ion is turned on, it is always observed to stay in bulk solution, and therefore the configuration space separation between these different λ windows results in the slow convergence of this approach.

The calculated relative binding free energies of all 31 charge-changing perturbations averaged over three independent FEP trials are plotted against experimental values in Figure 6. For

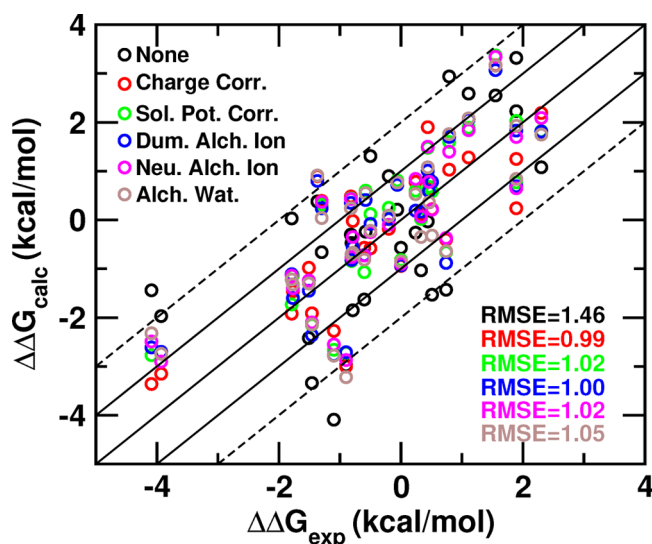


Figure 6. Comparison of calculated relative binding free energy with experiment. The mean of calculated relative binding free energies from three independent trials of FEP simulations are plotted against the experimental data. RMSEs were calculated using the mean calculated values versus experimental values. [Legend: black circles, uncorrected FEP; red circles, FEP with post-simulation charge correction; green circles, FEP with solvent potential correction; blue circles, FEP with dummy co-alchemical ion; magenta circle, FEP with neutral co-alchemical ion; and brown circles, FEP with co-alchemical water. Solid diagonal lines indicate the region within 1 kcal/mol of experimental values, while dashed diagonal lines indicate the region within 2 kcal/mol of experimental values.]

the uncorrected FEP (black circles in Figure 6), there are 11 charge-changing perturbations with errors of >1.5 kcal/mol. Of these 11 cases, 6 charge-changing perturbations have errors of >2 kcal/mol, and 5 have errors between 1.5 and 2 kcal/mol. The overall RMSE of 31 perturbations is 1.46 kcal/mol. In contrast, for the charge FEP methods (colored circles in Figure 6), the number of perturbations with errors of >1.5 kcal/mol are significantly reduced: 6 for dummy co-alchemical ion approach and 5 for the other two approaches. The resulting RMSEs ranges from 0.99 to 1.05 kcal/mol, significantly better than that observed with the uncorrected FEP.

Finally, we examined the convergence rate of the calculated free energies by plotting the RMSE as a function of simulation time (Figure 7). For the uncorrected FEP, the mean RMSE of three runs starts from 1.6 kcal/mol at 1 ns, then decreases to 1.5 kcal/mol and plateaus at 5 ns. In contrast, when the charge FEP methods are used, the mean RMSE starts from 1.3 to 1.5 kcal/mol at 1 ns, then is reduced to 1.1 kcal/mol and levels off

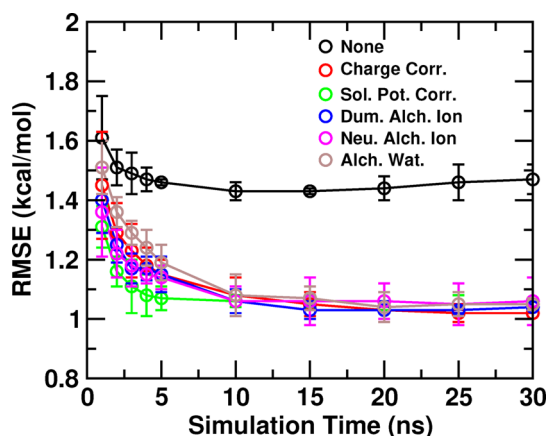


Figure 7. Convergence rate of the calculated relative binding free energies using various charge FEP protocols. The RMSE of calculated relative binding free energy is plotted as a function of simulation time. Mean RMSEs of three FEP simulations are shown, and the error bars are the standard deviation of three independent runs. [Legend: black, uncorrected FEP; red, FEP with post-simulation charge correction; green, FEP with solvent potential correction; blue, FEP with dummy co-alchemical ion; magenta, FEP with neutral co-alchemical ion; and brown, FEP with co-alchemical water.]

at 10 ns. The difference between the curves of the three charge FEP methods are within statistical error of each other, which suggests that the rates of convergence of the three approaches are roughly the same.

DISCUSSION

In this work, we have reported an implementation of three different charge FEP methods to calculate relative binding free energies between ligands with different net charges. With each of the three charge FEP methods, the box size dependence of calculated relative binding free energies associated with the brute force FEP method is effectively eliminated (see Figures 1–3). We further performed a large scale test of each of these methods on 31 charge-changing perturbations for 10 protein targets (Table 1). The results of the testing demonstrated that calculations with the charge FEP methods are quite accurate, with an RMSE of 1.1 kcal/mol (see Table 4 and Figure 6), which is on par with the accuracy for the perturbations without net charge change.⁸

The three charge FEP methods that we have implemented and tested here are post-simulation charge correction, solvent potential correction, and co-alchemical ion. The co-alchemical ion method includes three variants: dummy co-alchemical ion, neutral co-alchemical ion, and co-alchemical water. In our testing, these methods work equally well. Their accuracy are the same within statistical error (Table 4), and the rate of convergence of the methods also appear to be similar (Figure 7). While these methods give equally good results, each has its own advantages and disadvantages. The charge correction method in its current form requires assumptions that the simulation box is cubic and the environment surrounding the protein has the same dielectric constant. The former assumption can be relaxed when the lattice-sum integration constants for different shaped boxes are calculated numerically (see eq 2). The latter assumption is required for calculating the RIP term by solving the Poisson–Boltzmann equation (see eq 3). However, we notice that more-sophisticated Poisson–Boltzmann calculation can be done with heterogeneous

dielectric environment. For example, the APBSmem program⁵² was developed to solve the Poisson–Boltzmann equation in the presence of a membrane. The solvent potential correction method has the same limitations as the charge correction method. In comparison, the co-alchemical ion method does not require the assumptions about the shape of simulation box and the composition of the system, therefore it might be easier to generalize for more systems. Among the three variants of the co-alchemical ion method, some differences exist. For the dummy co-alchemical ion variant, we observed that, in the intermediate λ windows, co-alchemical ions can enter and remain in the hydrophobic protein interior, causing slow convergence and larger sampling errors (see Table 5). This problem is partially solved with the neutral co-alchemical ion variant, because the neutral form of a co-alchemical ion maintains van der Waals interactions with the environment and is less likely to enter the protein interior. However, for a membrane system, a neutral Lennard-Jones particle would very likely penetrate into a lipid membrane as well; thus, the neutral co-alchemical ion approach may have similar problems as the dummy co-alchemical ion approach for these systems. This issue was resolved with the use of co-alchemical water because a water molecule interacts stronger with water molecules themselves than with the membrane environment. Based on these results, we expect that the co-alchemical water variant of the co-alchemical ion method is the most general and robust approach for charge-changing perturbations among the methods we tested. Compared with post-simulation charge correction and solvent potential correction methods, the results from all three variants of the co-alchemical method were found to have larger standard deviations between different runs for some systems (Figures 1–3). This is probably due to the slow relaxation of the alchemical ion/water in the simulation box, and a biasing potential to prevent the alchemical ion from moving too close to the protein might help to alleviate this problem.

Ligands with titratable sites introduce additional complexity in the calculation of binding free energies. If the pK_a is close to pH in solvent, the ligand should be expected to have mixed populations of protonated and deprotonated forms. Furthermore, the dominant form of the ligands may differ upon binding to its target protein. To account for these situations, we have introduced the pK_a correction method (eq 7), similar to what has been employed by other researchers.^{53,54} Using IRAK4 as an example system, we have demonstrated that the pK_a correction method significantly improves calculated relative binding free energies (Table 3). In fact, in our test set, the calculated relative binding free energies of four charge-changing perturbations for three protein targets have a >1 kcal/mol improvement with the use of pK_a correction. Therefore, we suggest to use the pK_a correction method as a good practice for calculations involving ligands with titratable sites. Although the pK_a method was only used to account for a single titratable site in the current study, it can be easily extended for multiple uncoupled titratable sites. For general cases with multiple coupled titratable sites, the Wyman linkage relationship⁵⁵ can be used in combination with constant pH MD simulations to compute pH-dependent binding free energies.^{53,54,56}

An interesting finding of this work is the important role of the ionic strength of buffer solution on the calculated relative binding free energy. With addition of salts, we observed significant changes of calculated relative binding free energies

in two cases for PTP1B and DLK (Figures 1 and 2), but not in the other case for CDK2 (Figure 3). In the case of PTP1B, charged ligand 26 becomes much more potent with the addition of salts compared to simulations with only counterions (Figure 1). This is mainly due to an ion-mediated interaction between the protein and the charged ligand. In particular, the charged piperidine group of ligand 26 and the two nearby arginines in the protein binding pocket are bridged by a Cl^- ion in the simulations with added salts, reducing the repulsive interactions between ligand 26 and the arginines. In contrast, for DLK, charged ligand 4 becomes less potent with additional salts (Figure 2). In this case, the salt screening effects weaken the attractive interactions between the charged piperidine group of ligand 4 and nearby Glu200 of the protein, resulting in the reduced binding potency, compared to simulations without salts. In the third case of CDK2, the calculated relative binding free energy does not change upon adding salts (see Figure 3). We speculate that the salt screening effects on the interactions between charged ligand 39 and the two opposite charged protein residues (Lys89 and Asp86) compensate each other, resulting in the binding being insensitive to salt concentration. These examples demonstrate that the effect of salts on ligand binding is complex and system-specific: it can enhance or reduce the binding potency for charged ligand, as well as minimally impact the binding in certain cases. The same phenomenon was observed in previous studies as well. For example, in the protein–ligand binding affinity experiment that was reported by Papanephoytou et al.,²³ they found that the binding affinity is not affected by salt concentration for one system while the salt concentration has significant but opposite effects for the other two systems. They further suggested that the different effects of salts on these three systems are related to the different hydrophobic or hydrophilic nature of the binding site and ligand. Considering the important role of ionic strength in the relative binding free energies of charge-changing perturbations, we suggest to perform charge FEP simulations with the same ionic strength as the experiment. We note that it is not trivial to determine the number of ions to add in a finite simulation box to match the experimental assay condition. As noted by Schmit et al.,⁵⁷ simply multiplying the salt concentration by the solvent volume in the simulation box may result in a higher effective salt concentration than intended. The Grand Canonical Monte Carlo method proposed by Ross et al.⁵⁸ might be a better way to model ions, which is subject to future studies.

While the charge FEP methods give excellent results for relative binding free energies for charge-changing perturbations for the cases tested in this paper, there are also some limitations. First, compared with neutral perturbations, we observed larger variations between different trials of simulations for the same charge-changing perturbation. We suspect this is due to the slow sampling of ion configurations and charge–charge interactions. When the salt concentration is low, extensive simulation is required to reach a converged ion distribution. In addition, if ions can form specific interactions in the binding site such as the case of PTP1B, the fraction of simulation time manifesting these interactions may vary significantly between different runs. Furthermore, if a charged group of a ligand interacts with a charged protein residue, it may be difficult to obtain converged sampling of salt bridge formation and breaking.⁵⁹ The net-charge change of the ligand may cause reorganization of protein residues as well, which may be difficult to sample through brute force MD. Therefore,

FEP simulations for charge-changing perturbations may require longer time to converge than neutral perturbations. According to our testing, at least 10 ns per λ window is generally required (Figure 7). Another limitation of the current methods is that, when a charged group being perturbed is in a buried pocket, the relative binding free energy may be overestimated by the calculations. For these cases, the direction of the calculated binding free energy change is usually correct, but the magnitude is significantly larger than the experimental value. Similar problems were also reported in previous studies. For example, Wu and Brooks⁶⁰ calculated the pK_a values of buried ionizable residues in the staphylococcal nuclease variants,^{61,62} and their calculated pK_a shifts, which are related to the free-energy changes between protonated and deprotonated forms of a residue, are significantly overestimated, compared to the experiment. After they placed hydronium ions near the acidic residue, they obtained comparable pK_a values as the experiment. Therefore, they hypothesize that a released proton from deprotonation may remain near the deprotonated residue inside the dehydrated pocket. The relatively small pK_a shifts for buried ionizable residues correspond to high apparent dielectric constants in the protein interior. Other explanations for the high apparent dielectric constants have been proposed previously, including water penetration⁶³ and conformational relaxation.⁶⁴ These may also explain why we overestimated the relative binding free energy for a charge-changing perturbation in deeply buried pockets.

In summary, we have implemented three reported charge FEP methods to calculate relative binding free energies for charge-changing perturbations to facilitate comparative testing. The methods effectively eliminate the box size dependence of the calculated binding free energy associated with the brute force FEP method. Tested on 10 protein targets and 31 charge-changing perturbations, the methods yielded a high level of accuracy, with an RMSE of 1.1 kcal/mol between the calculated binding free energies and experimental data. These methods open a new avenue for structure-based drug discovery, where modifications changing the ligand net charge are often used to improve binding potency, solubility, and other ADME/Tox properties. In addition, the charge FEP methods can also be used to calculate the change of protein–protein binding free energy upon mutation of charged residues, which is critical for biologics drug discovery, such as antibody drugs.

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ACKNOWLEDGMENTS

We thank Anthony Clark and Jennifer L. Knight for their helpful discussion and suggestions for the development of the charge FEP methods.

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